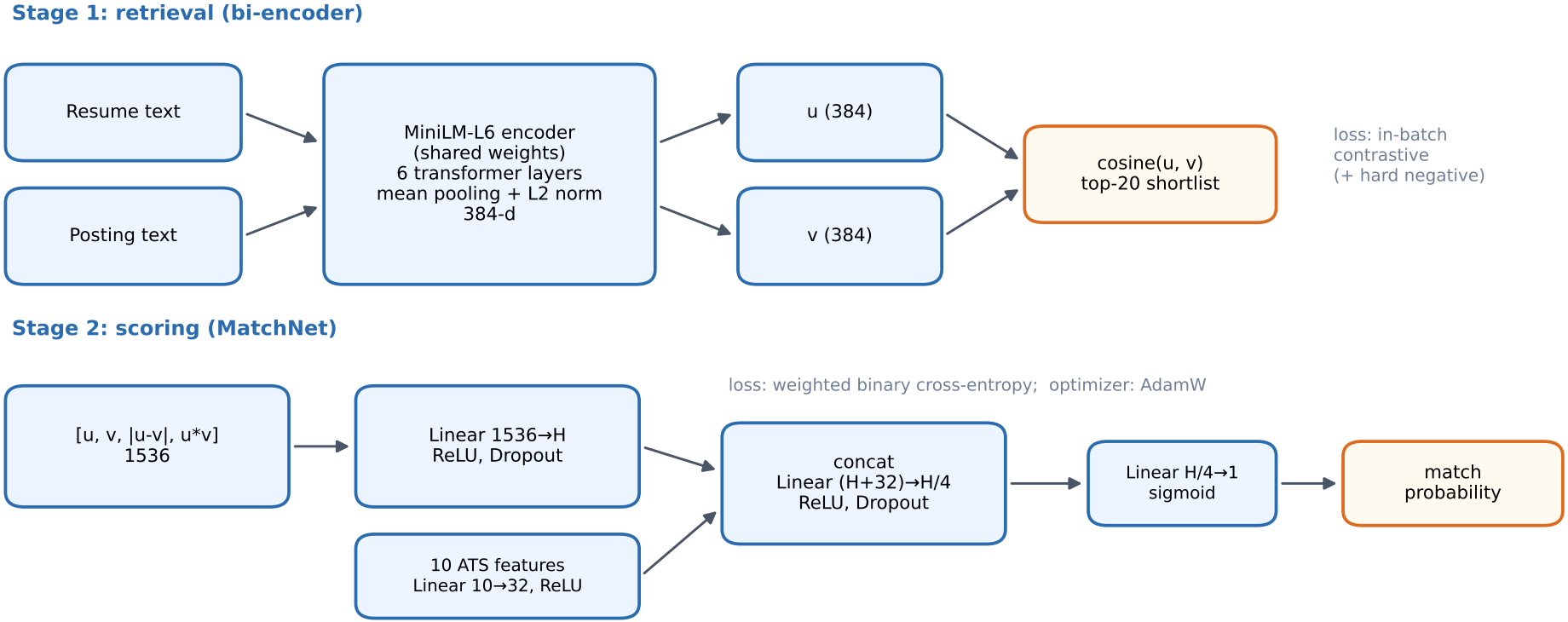
Stage 1: retrieval (bi-encoder)
Resume text
Posting text
MiniLM-L6 encoder
(shared weights)
6 transformer layers
mean pooling + L2 norm
384-d
u (384)
v (384)
cosine(u, v)
top-20 shortlist
loss: in-batch
contrastive
(+ hard negative)
Stage 2: scoring (MatchNet)
loss: weighted binary cross-entropy; optimizer: AdamW
[u, v, |u-v|, u*v]
1536
Linear 1536→H
ReLU, Dropout
10 ATS features
Linear 10→32, ReLU
concat
Linear (H+32)→H/4
ReLU, Dropout
Linear H/4→1
sigmoid
match
probability